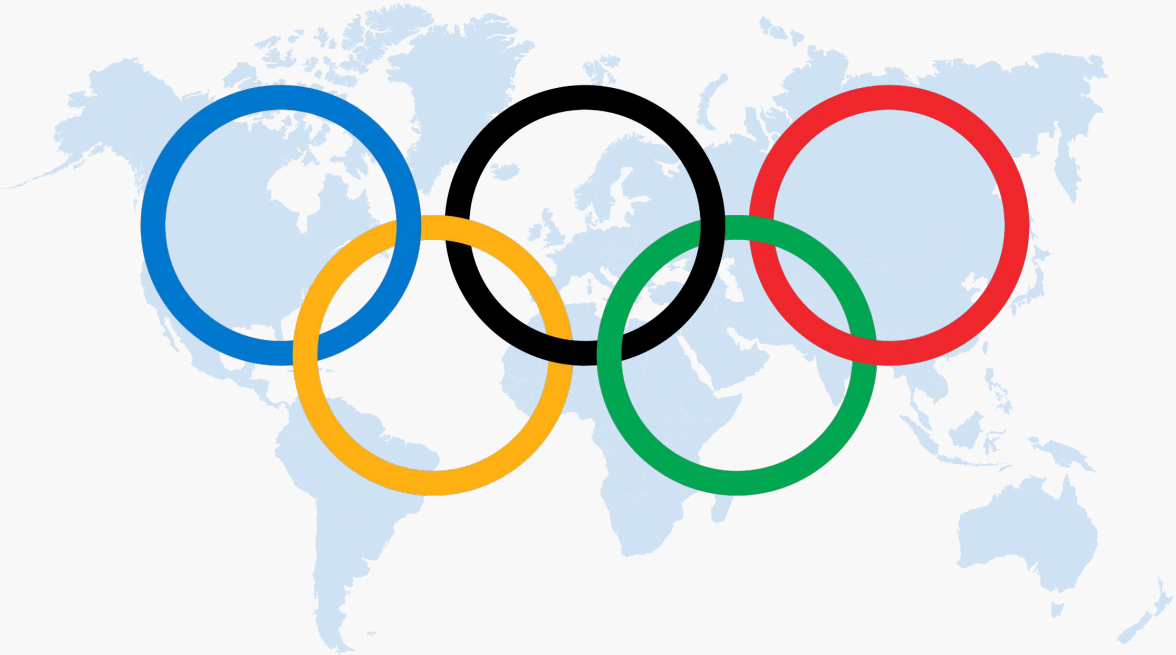
OLYMPICS DATA ANALYSIS

THE OLYMPIC SYSTEM AS A MIRROR OF GLOBAL PROGRESS



Bekka Ayele DEBEBE
Mikiko NOMIYA

Do the Olympics show a trend of Global Progress ?

1. **The Economic Pillar: Financial Decoupling**

   Is a nation's wealth connected to its performance in the Olympics?

2. **The Social Pillar: Gender Parity**

   Is the Olympic system becoming more gender inclusive?

3. **The Biological Pillar: Athletes' Performance Longevity**

   Do recent athlete age trends reflect the evolution of modern sports medicine?

The Datasets

We used a dataset of **athletes** that participated in the games and a dataset of the **GDP** of nations over the years.
We cleaned and modified the datasets.
The process included filtering, feature engineering and merging of the datasets to meet our needs.

Links

- Athletes' Dataset
- GDP Dataset

ID	Name	Sex	Age	Height	Weight	Team	NOC	Games	Year	Season	City	Sport	Event	Medal
1	A Dijiang	M	24	180	80	China	CHN	1992 Summer	1992	Summer	Barcelona	Basketball	Basketball Men's	NA
2	A Lamusi	M	23	170	60	China	CHN	2012 Summer	2012	Summer	London	Judo	Judo Men's Extra	NA
3	Gunnar Nielsen A	M	24	NA	NA	Denmark	DEN	1920 Summer	1920	Summer	Antwerpen	Football	Football Men's F	NA
4	Edgar Lindenau A	M	34	NA	NA	Denmark/Swede	DEN	1900 Summer	1900	Summer	Paris	Tug-Of-War	Tug-Of-War Mer	Gold
5	Christine Jacoba	F	21	185	82	Netherlands	NED	1988 Winter	1988	Winter	Calgary	Speed Skating	Speed Skating V	NA
5	Christine Jacoba	F	21	185	82	Netherlands	NED	1988 Winter	1988	Winter	Calgary	Speed Skating	Speed Skating V	NA
5	Christine Jacoba	F	25	185	82	Netherlands	NED	1992 Winter	1992	Winter	Albertville	Speed Skating	Speed Skating V	NA
5	Christine Jacoba	F	25	185	82	Netherlands	NED	1992 Winter	1992	Winter	Albertville	Speed Skating	Speed Skating V	NA
5	Christine Jacoba	F	27	185	82	Netherlands	NED	1994 Winter	1994	Winter	Lillehammer	Speed Skating	Speed Skating V	NA
5	Christine Jacoba	F	27	185	82	Netherlands	NED	1994 Winter	1994	Winter	Lillehammer	Speed Skating	Speed Skating V	NA
6	Per Knut Aaland	M	31	188	75	United States	USA	1992 Winter	1992	Winter	Albertville	Cross Country S	Cross Country S	NA
6	Per Knut Aaland	M	31	188	75	United States	USA	1992 Winter	1992	Winter	Albertville	Cross Country S	Cross Country S	NA
6	Per Knut Aaland	M	31	188	75	United States	USA	1992 Winter	1992	Winter	Albertville	Cross Country S	Cross Country S	NA
6	Per Knut Aaland	M	31	188	75	United States	USA	1992 Winter	1992	Winter	Albertville	Cross Country S	Cross Country S	NA
6	Per Knut Aaland	M	33	188	75	United States	USA	1994 Winter	1994	Winter	Lillehammer	Cross Country S	Cross Country S	NA
6	Per Knut Aaland	M	33	188	75	United States	USA	1994 Winter	1994	Winter	Lillehammer	Cross Country S	Cross Country S	NA
6	Per Knut Aaland	M	33	188	75	United States	USA	1994 Winter	1994	Winter	Lillehammer	Cross Country S	Cross Country S	NA
6	Per Knut Aaland	M	33	188	75	United States	USA	1994 Winter	1994	Winter	Lillehammer	Cross Country S	Cross Country S	NA
7	John Aalberg	M	31	183	72	United States	USA	1992 Winter	1992	Winter	Albertville	Cross Country S	Cross Country S	NA
7	John Aalberg	M	31	183	72	United States	USA	1992 Winter	1992	Winter	Albertville	Cross Country S	Cross Country S	NA
7	John Aalberg	M	31	183	72	United States	USA	1992 Winter	1992	Winter	Albertville	Cross Country S	Cross Country S	NA

Country	Country Co	1960	1961	1962	1963	1964	1965	1966
Malaysia	MYS	1.92E+09	1.9E+09	2E+09	2.51E+09	2.67E+09	2.96E+09	3.14E+09
North Ame	NAC	5.84E+11	6.04E+11	6.47E+11	6.84E+11	7.35E+11	7.98E+11	8.76E+11
Namibia	NAM							
New Calec	NCL						1.6E+08	1.64E+08
Niger	NER	4.5E+08	4.86E+08	5.32E+08	5.86E+08	5.83E+08	6.73E+08	7.02E+08
Nigeria	NGA	4.2E+09	4.47E+09	4.91E+09	5.17E+09	5.55E+09	5.87E+09	6.37E+09
Nicaragua	NIC	2.27E+08	2.44E+08	2.69E+08	2.97E+08	3.47E+08	5.64E+08	6.07E+08
Netherlan	NLD	1.23E+10	1.35E+10	1.46E+10	1.59E+10	1.87E+10	2.1E+10	2.29E+10
Norway	NOR	5.16E+09	5.63E+09	6.07E+09	6.51E+09	7.16E+09	8.06E+09	8.7E+09
Nepal	NPL	5.08E+08	5.32E+08	5.74E+08	4.97E+08	4.96E+08	7.35E+08	9.07E+08
Nauru	NRU							
New Zeala	NZL	5.49E+09	5.67E+09	6.08E+09	6.64E+09	7.27E+09	5.65E+09	5.86E+09

Some Custom Datasets

(Made from the other two)

NOC	Year	Season	Gold	Silver	Bronze	Total_Medals	Sports_List
AFG	2008	Summer	0	0	1	1	['Taekwondo']
AFG	2012	Summer	0	0	1	1	['Taekwondo']
AHO	1988	Summer	0	1	0	1	['Sailing']
ALG	1984	Summer	0	0	2	2	['Boxing']
ALG	1992	Summer	1	0	1	2	['Athletics', 'Boxing']
ALG	1996	Summer	2	0	1	3	['Athletics', 'Boxing']
ALG	2000	Summer	1	1	3	5	['Athletics', 'Boxing']
ALG	2008	Summer	0	1	1	2	['Judo']
ALG	2012	Summer	1	0	0	1	['Athletics']
ALG	2016	Summer	0	2	0	2	['Athletics']

Nations performance

The number of medals won by a nation and the sports they won those medals in.

ISO_Code	Period	Dominance	Country Code	GDP
ARG	1960-1968	0.07676551	ARG	2.49E+10
AUS	1960-1968	0.14635674	AUS	2.34E+10
KOR	1960-1968	0.0786636	KOR	3.57E+09
FRA	1960-1968	0.24197412	FRA	8.93E+10
TUN	1960-1968	0.07209835	TUN	1.04E+09
IRL	1960-1968	0.03570087	IRL	2.61E+09
FIN	1960-1968	0.17086905	FIN	7.41E+09
NGA	1960-1968	0.03570087	NGA	5.22E+09
GBR	1960-1968	0.20946059	GBR	9.21E+10
USA	1960-1968	0.33387481	USA	6.82E+11
JPN	1960-1968	0.19736994	JPN	7.88E+10
GHA	1960-1968	0.03570087	GHA	1.64E+09

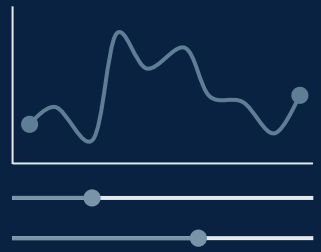
GDP and Dominance

A nation's average GDP and its *Dominance in the olympic games over an 8 year period.

*Dominance will be explained later.

01 The Economic Pillar

- ***Does wealth have an effect on the performance of a nation and, if so, is the effect reducing?***



The Approach

01

Define “Dominance”

Build a network of nations and have their Eigenvector Centrality as their “Dominance”

02

Compare to GDP

Check whether there is a correlation between GDP and Dominance using Spearman Rank Correlation

03

Check for progress

Visualize the change of the correlation over time

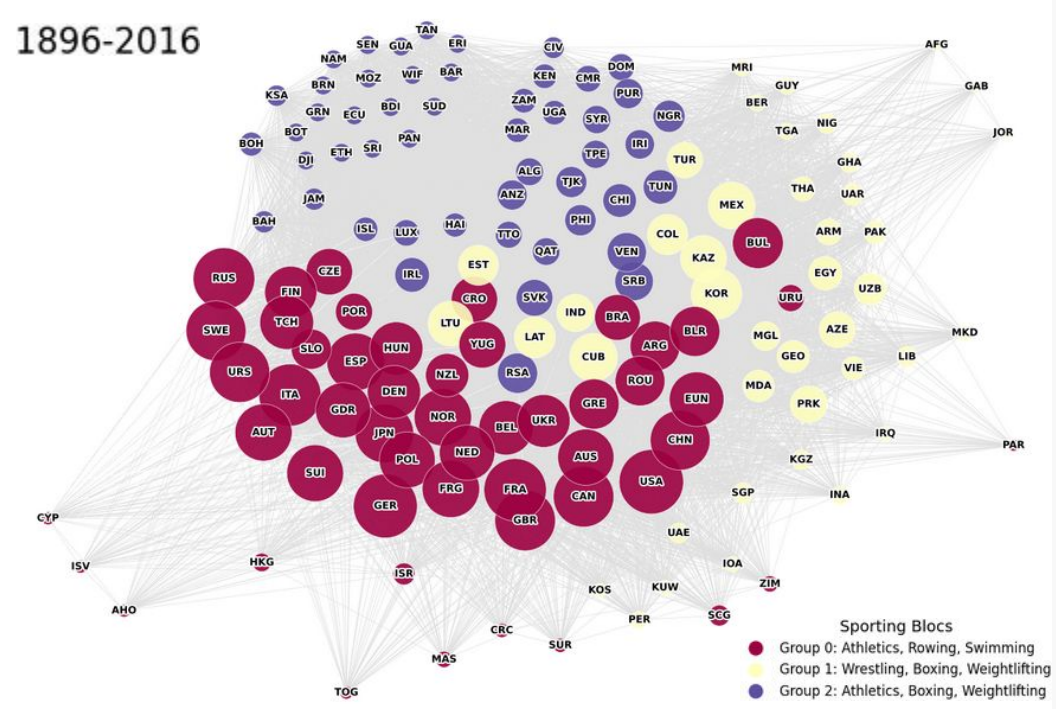
The Network

Bipartite Foundation

Start with a bipartite network of nations and the sports they succeeded in.

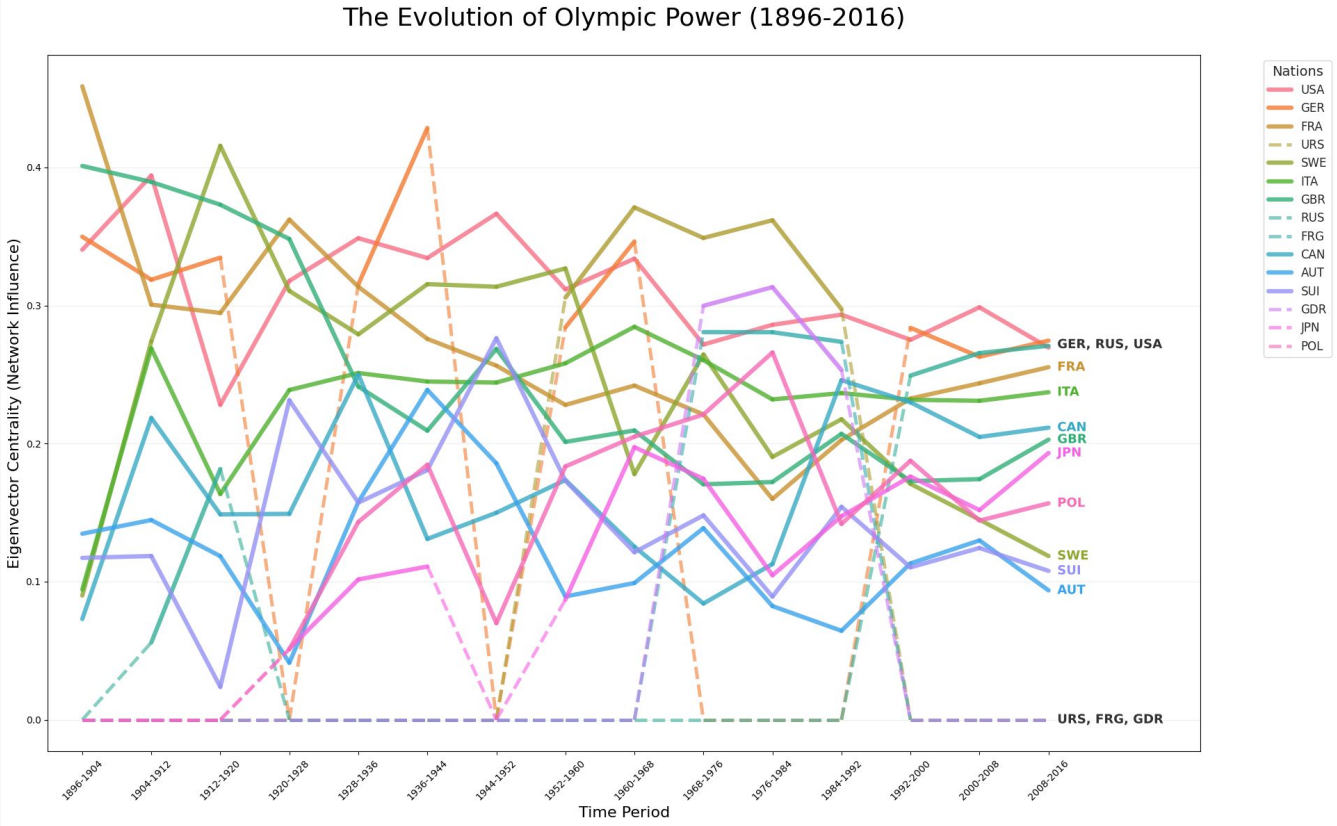
Nation-to-Nation Projection

A connection between nations exists if they both won a medal in the same sport. The **Eigenvector Centrality** of a nation is its **Dominance**



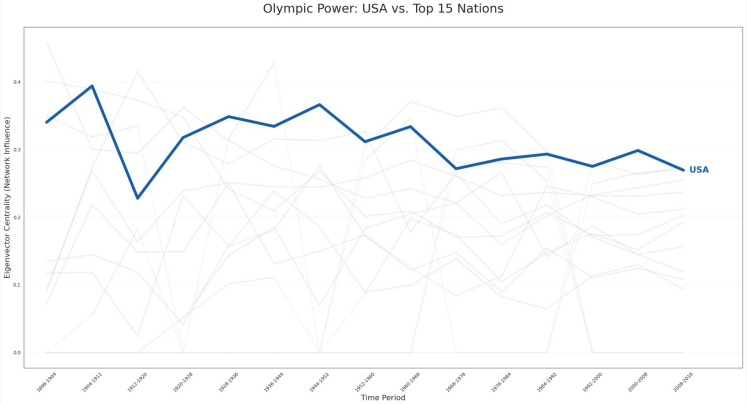
Note: MNE, LIE, FIJ, NEP, MON have been detached for the visualization.

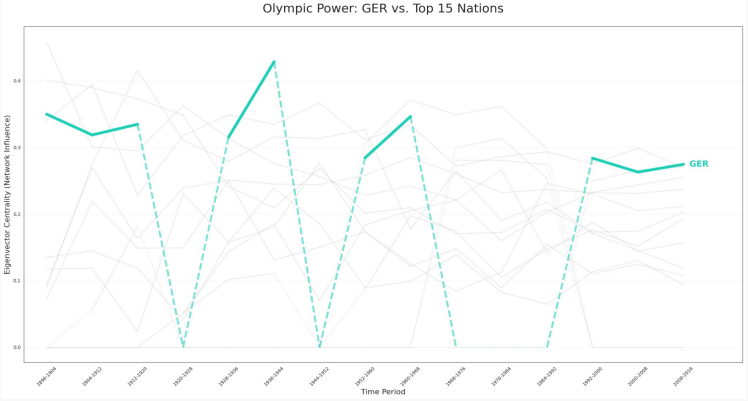
Dominance of Nations Over Time

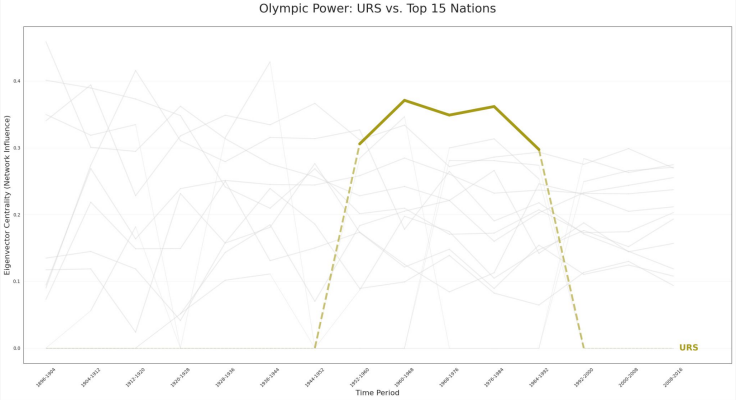


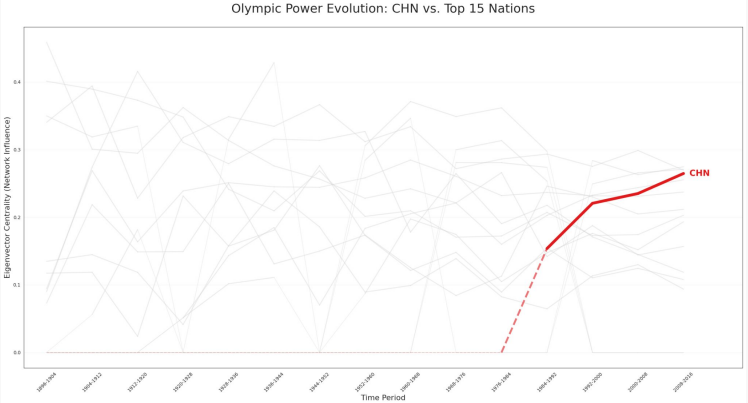
Shows the shifts in power of the top 15 most dominant nations by plotting the dominance of the nations in different time windows.

Trends of Individual Nations





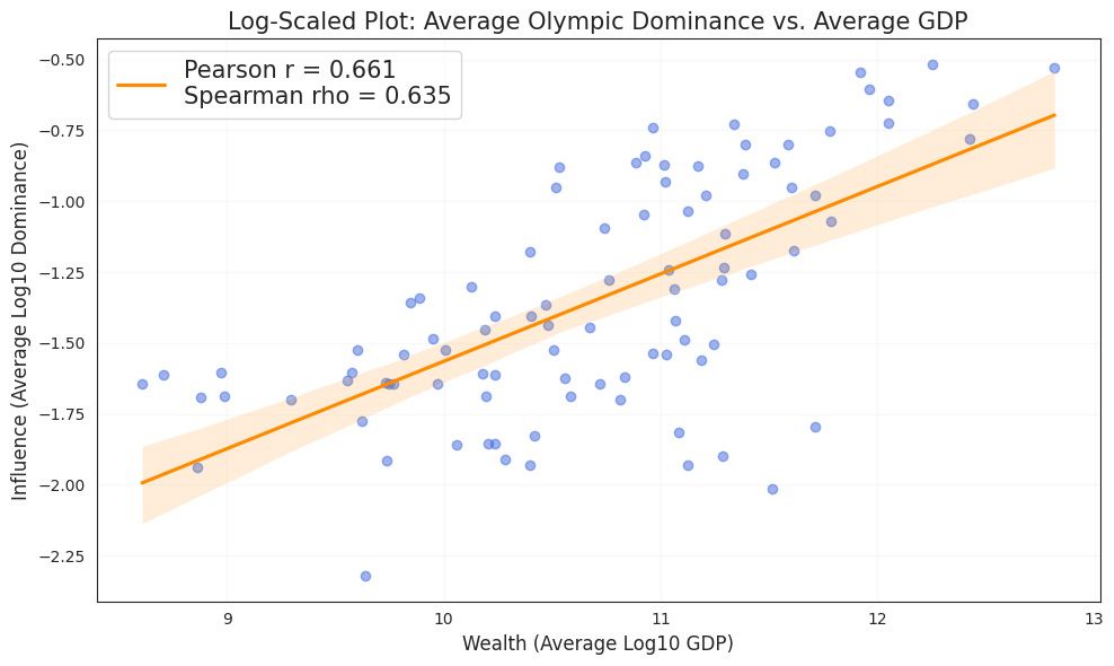




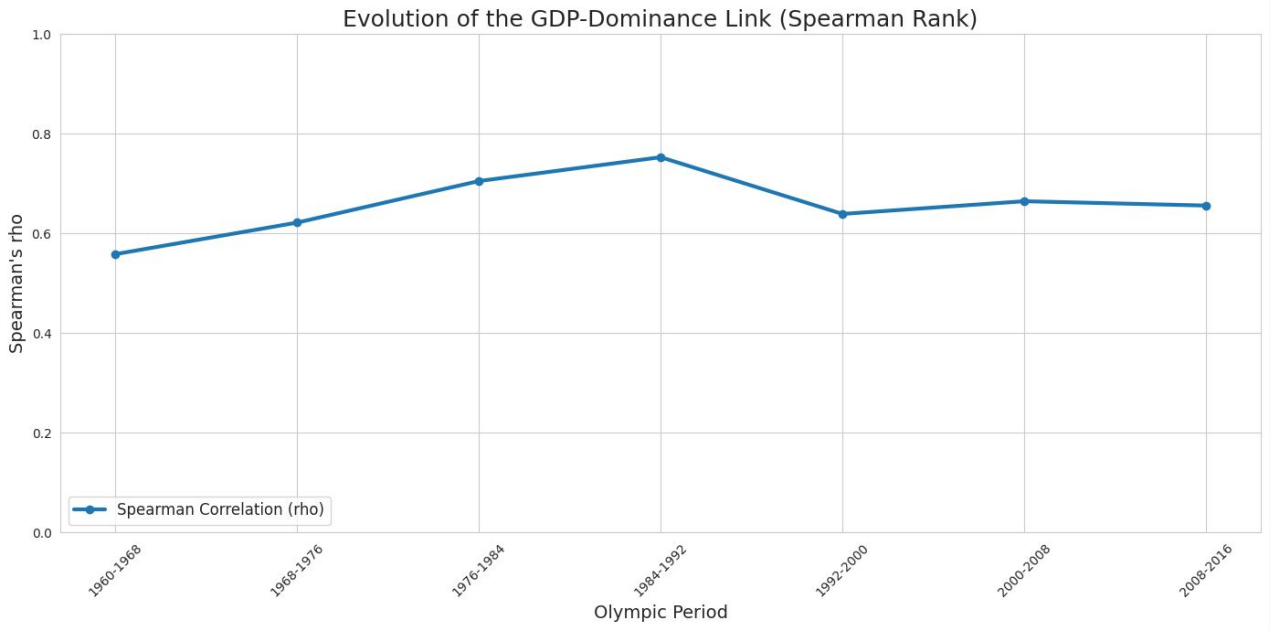
Correlation of GDP and Dominance

The **Spearman Rank Correlation Coefficient** for the average GDP and the average Dominance of each country shows a **moderate positive correlation.**

```
Spearman Correlation (rho): 0.635
Spearman P-Value: 8.036e-12
```



The Correlation Over Time



Conclusion

The Correlation has more or less stayed the same over the past 20 years and GDP and Olympic Dominance remain linked

--- SPEARMAN CORRELATION BY ERA ---

	Period	N_Countries	Spearman_Rho	Spearman_P
0	1960-1968	36	5.5844e-01	4.0090e-04
1	1968-1976	36	6.2176e-01	5.1737e-05
2	1976-1984	34	7.0515e-01	3.2160e-06
3	1984-1992	52	7.5302e-01	1.1980e-10
4	1992-2000	70	6.3919e-01	2.5978e-09
5	2000-2008	69	6.6471e-01	4.7145e-10
6	2008-2016	76	6.5600e-01	1.2581e-10

02 The Social Pillar

- ***How has the absolute and proportional participation of male and female athletes evolved across Summer and Winter Games?***

Data preprocessing

Temporal Binning: we **grouped** consecutive olympic years into **blocks of 2** (separately for Summer and Winter)

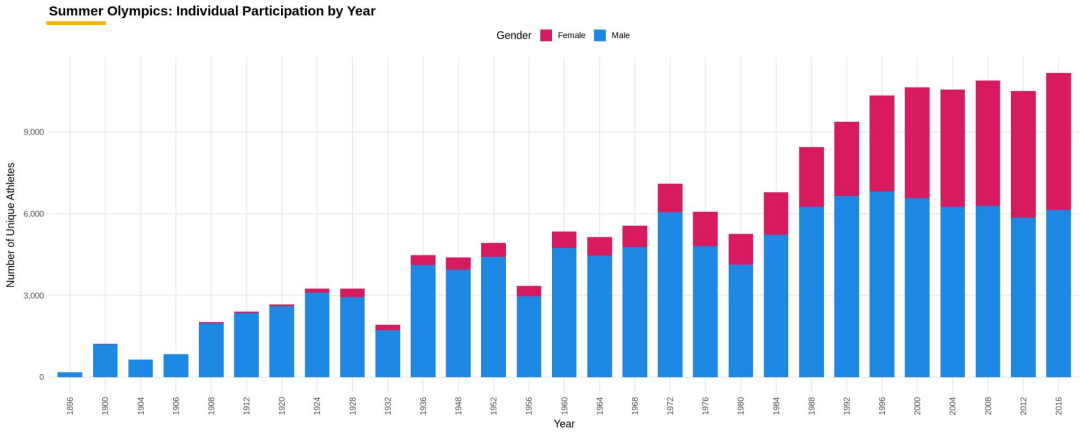
Methodology

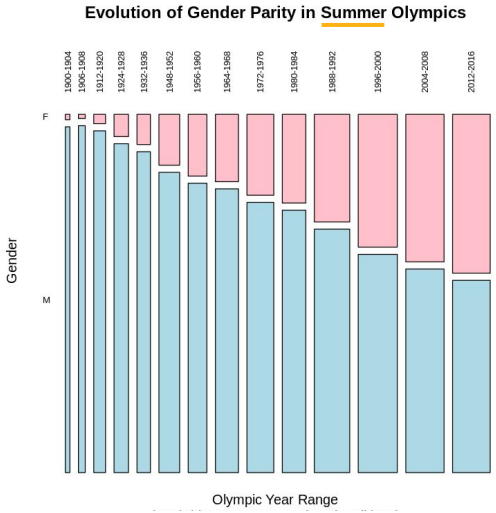
We performed **Chi-Squared Tests of Independence** on Summer and Winter **contingency tables** to determine if the shift in gender distribution is statistically significant

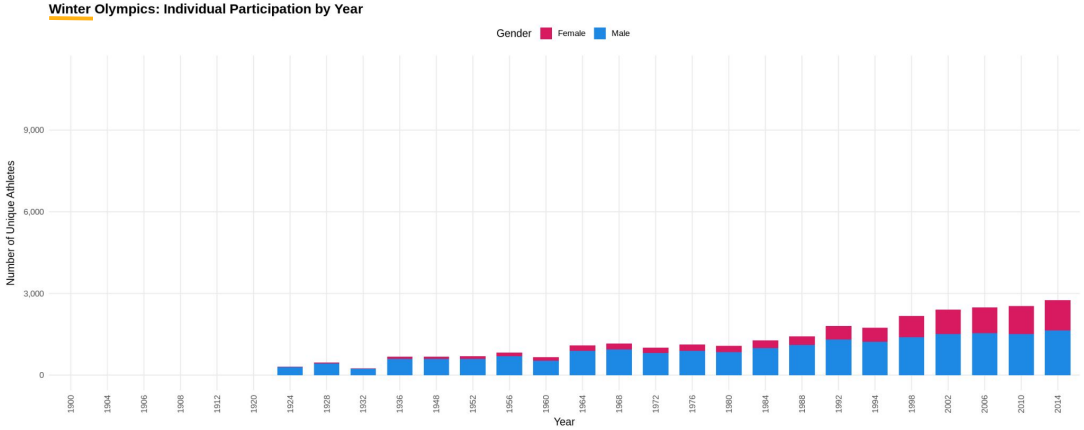
Observed Frequencies:

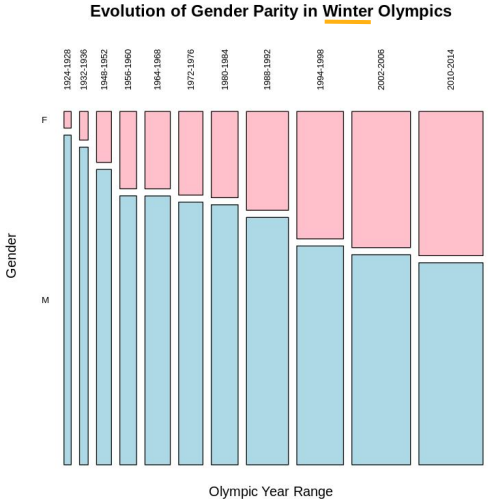
```
[1] "Contingency Table (Winter):"

                F    M
  1924-1928    50  992
  1932-1936   103 1144
  1948-1952   318 1845
  1956-1960   541 1882
  1964-1968   820 2849
  1972-1976   849 2667
  1980-1984   966 2914
  1988-1992  1734 4341
  1994-1998  2489 4276
  2002-2006  3339 5152
  2010-2014  3870 5423
```

```
[1] "Contingency Table (Summer):"

                 F     M
  1900-1904     49  3188
  1906-1908     58  4776
  1912-1920    221  8111
  1924-1928    648  9577
  1932-1936    815  8660
  1948-1952   2125 12550
  1956-1960   2328 10918
  1964-1968   3125 13165
  1972-1976   4365 14580
  1980-1984   4203 12442
  1988-1992   7667 17347
  1996-2000  10439 17162
  2004-2008  11362 15683
  2012-2016  12038 14570
```


Summer Olympics: Individual Participation by Year
Gender
Female
Male
Number of Unique Athletes
0
3,000
6,000
9,000
1896
1900
1904
1906
1908
1912
1920
1924
1928
1932
1936
1948
1952
1956
1960
1964
1968
1972
1976
1980
1984
1988
1992
1996
2000
2004
2008
2012
2016
Year


Evolution of Gender Parity in Summer Olympics
1900-1904
1906-1908
1912-1920
1924-1928
1932-1936
1948-1952
1956-1960
1964-1968
1972-1976
1980-1984
1988-1992
1996-2000
2004-2008
2012-2016
F
M
Gender
Olympic Year Range


Winter Olympics: Individual Participation by Year
Gender
Female
Male
Number of Unique Athletes
0
3,000
6,000
9,000
1900
1904
1906
1908
1912
1920
1924
1928
1932
1936
1948
1952
1956
1960
1964
1968
1972
1976
1980
1984
1988
1992
1994
1998
2002
2006
2010
2014
Year


Evolution of Gender Parity in Winter Olympics
1924-1928
1932-1936
1948-1952
1956-1960
1964-1968
1972-1976
1980-1984
1988-1992
1994-1998
2002-2006
2010-2014
F
M
Gender
Olympic Year Range

Results and Interpretation

Null Hypothesis (H0): Gender and Olympic year groups are independent

```
[1] "Chi-squared test result (Summer):"

        Pearson's Chi-squared test

data:  summer_table_grouped
X-squared = 21000, df = 13, p-value < 2.2e-16
```

```
[1] "Chi-squared test result (Winter):"

        Pearson's Chi-squared test

data:  winter_table_grouped
X-squared = 2151.4, df = 10, p-value < 2.2e-16
```

The **Pearson's Chi-squared tests** for both Summer and Winter Olympics yielded **extremely high Chi-squared values (X^2)**, resulting in **small p-values** (significantly less than 0.05).

Conclusion: We **reject** the null hypothesis.

There is a **statistically significant association** between gender and the Olympic year groups for both Summer and Winter Games.

03 The Biological Pillar

- ***Do recent athlete age trends reflect the evolution of modern sports medicine?***

The Approach

1 Data cleaning

We deleted entries with missing Age values

2 Feature Engineering

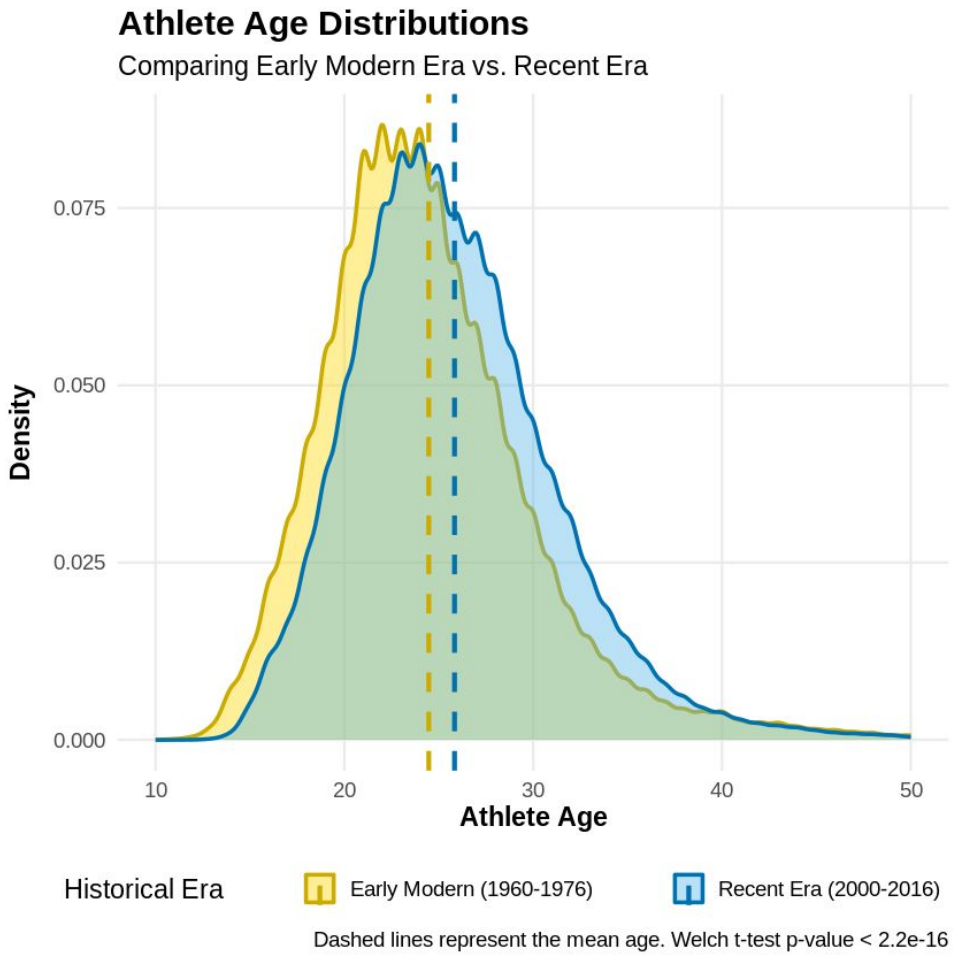
We performed a temporal grouping by defining two eras: The **Early Modern Era** (**1960-1976**) and The **Recent Era** (**2000-2016**)

3 Welch Two-Sample T-test

To determine if the difference in **mean athlete age** between these two eras was statistically significant

<u>Why?</u>

- Does not assume equal variances
- Given the massive sample size (around 130k), the distribution of the mean is normal



The **density plot** with Blue (Recent) and Yellow (Early) scales shows the rightward shift of the entire age distribution

Results

- **+1.36 years** in the average athlete's age
- **Test Statistic** (**t = 43.279**): the high t-value indicates that the observed difference is very large relative to the variation within the age data
- The **p-value** (**<2.2e-16**) is much lower than the standard level of 0.05, so we **reject** the Null Hypothesis

```
        Welch Two Sample t-test

data:  recent_era$Age and early_era$Age
t = 43.279, df = 103224, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 1.302595 1.426173
sample estimates:
mean of x mean of y
 25.83422  24.46983
```

Conclusion

Persistent Financial Links

A nation's wealth has a significant effect on its olympic success and there are no clear signs of this link fading.

Evolving Gender Parity

Female involvement has grown from nearly non-existent in the early 20th century to approaching parity in the modern era

Rising Athlete Age

This rightward shift in age distribution is strongly associated with the global evolution of sports medicine and recovery standards

THANK YOU